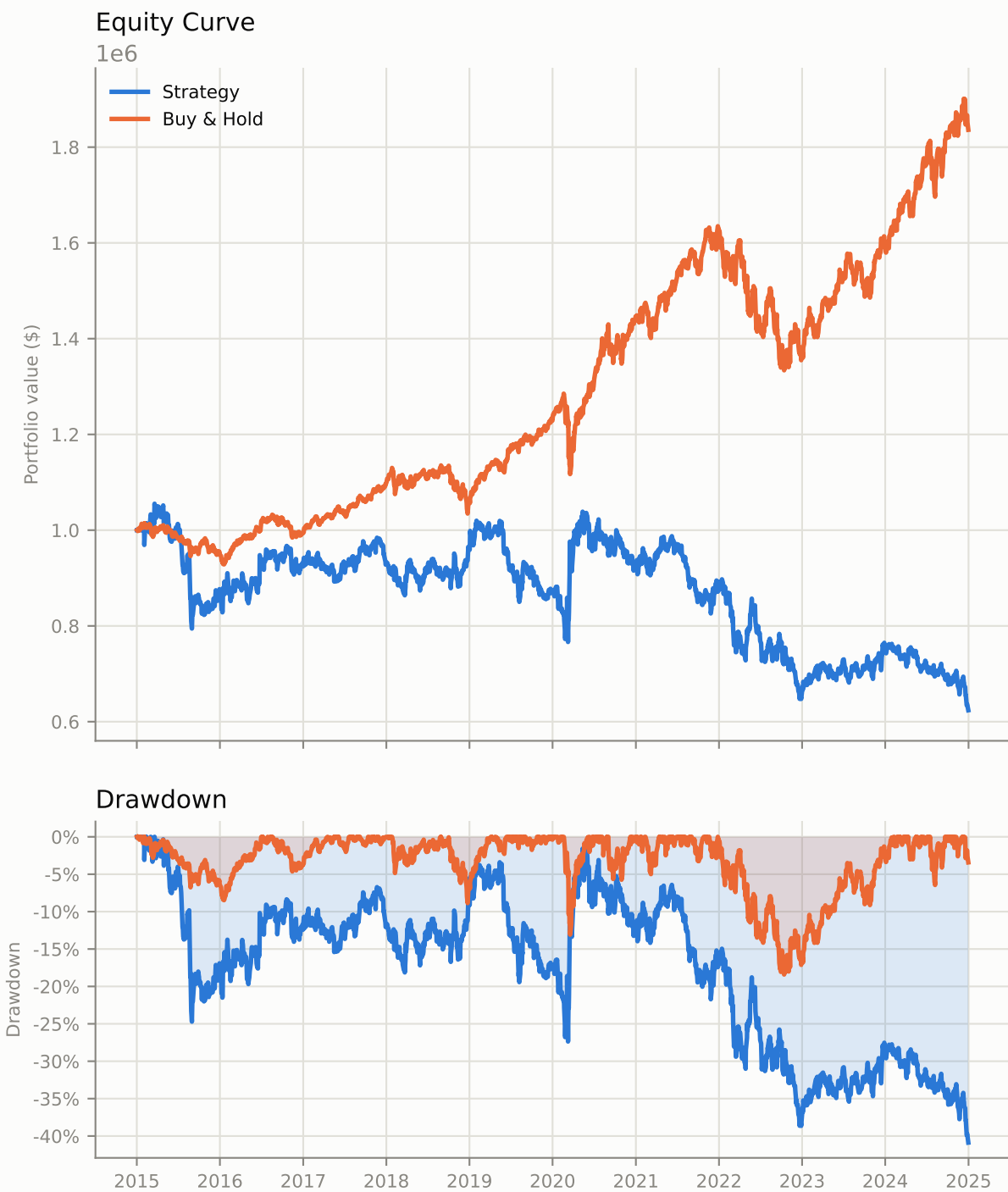


Performance Overview - Score Scaling Neutral



Metric	Strategy	Buy & Hold
Total return	-37.61%	83.62%
Annualized return	-4.61%	6.27%
Annualized vol	14.96%	8.55%
Sharpe	-0.31	0.73
Max drawdown	-40.88%	-18.41%
Drawdown/vol	2.73	2.15
Num trades	14,998	8
Win rate	48.93%	62.50%
Avg win	1,221.93	183,135.61
Avg loss	-1,219.72	-26,577.97
Risk/reward ratio	1.00	6.89
Payoff factor	0.96	11.48
CPC index	0.47	49.46
Time in market	99.17%	99.96%
Turnover	6884.62	2.21

Return Correlation

Strategy	1.00	0.06
Buy & Hold	0.06	1.00
	Strategy	Buy & Hold

Monthly Returns

Strategy

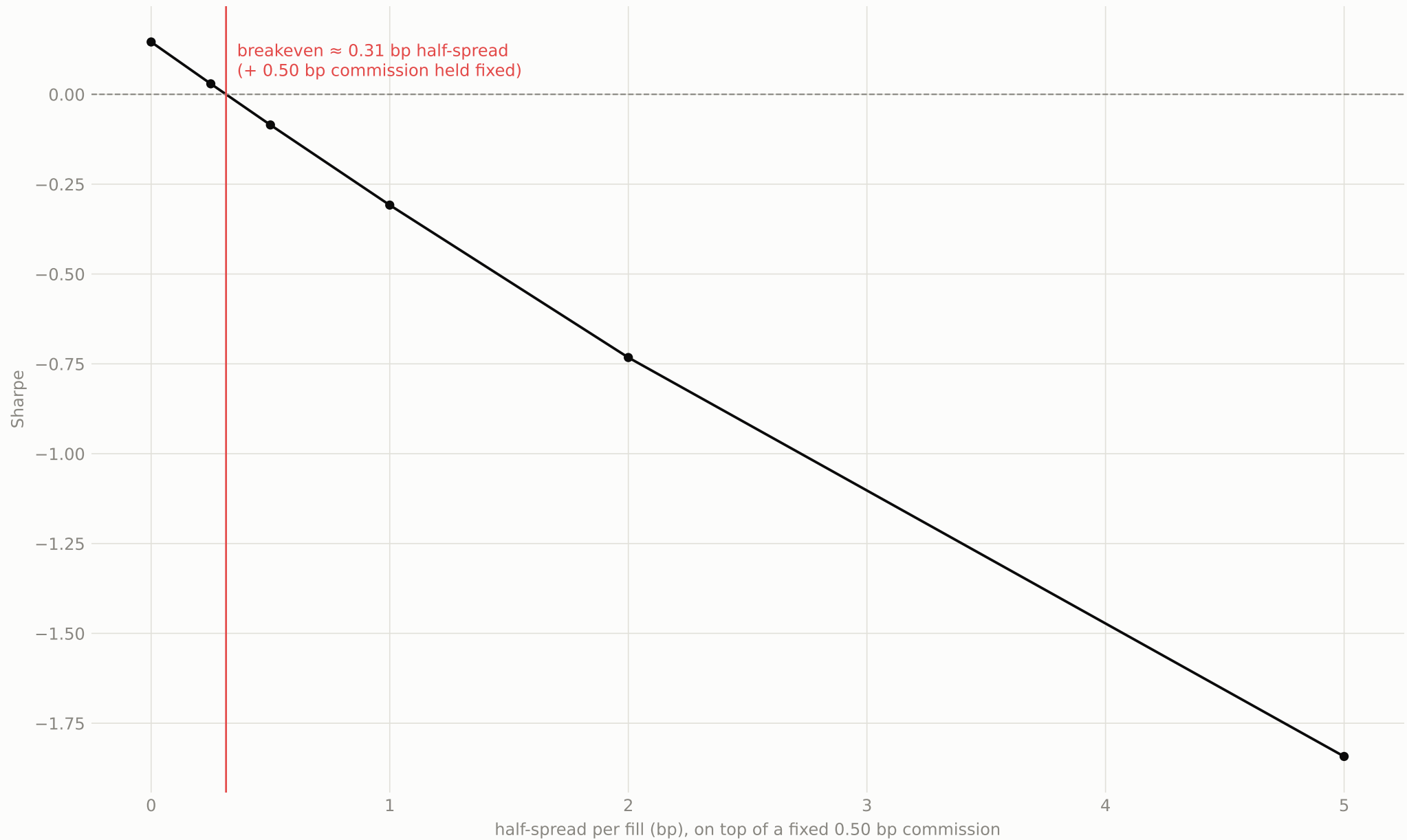
2015	0.0%	0.9%	4.0%	-3.4%	-3.7%	3.6%	-9.5%	-13.3%	8.2%	-2.5%	-0.3%	4.1%	-13.0%	-24.7%	-0.83
2016	2.0%	-1.1%	1.1%	-3.6%	4.6%	2.8%	2.5%	0.5%	0.5%	0.8%	-1.9%	-1.7%	7.7%	-6.9%	0.52
2017	0.7%	-0.1%	-0.9%	-1.1%	-1.8%	3.4%	0.3%	1.0%	1.9%	1.2%	0.8%	-4.9%	-0.9%	-6.4%	-0.10
2018	-1.4%	-2.5%	0.1%	2.3%	-2.8%	2.5%	2.5%	-2.6%	0.4%	1.7%	-1.3%	4.7%	3.2%	-7.3%	0.24
2019	5.3%	-0.4%	0.0%	-0.6%	-3.4%	-4.9%	-1.8%	2.3%	-0.8%	-4.5%	-1.3%	0.8%	-9.3%	-16.6%	-0.72
2020	-2.0%	-9.1%	23.0%	5.2%	2.9%	-5.1%	1.6%	-3.3%	-0.9%	2.1%	-4.5%	-0.7%	6.2%	-13.1%	0.27
2021	2.7%	-2.5%	3.0%	2.8%	-0.9%	-0.4%	-2.3%	-5.8%	-4.1%	1.3%	-1.1%	2.5%	-5.7%	-16.3%	-0.46
2022	-1.8%	-3.4%	-6.3%	-3.0%	9.5%	-6.9%	-3.5%	0.0%	2.1%	-3.8%	-3.6%	-6.7%	-26.4%	-27.8%	-1.43
2023	5.9%	2.0%	1.8%	-4.0%	-0.9%	6.2%	-5.0%	2.7%	-0.2%	1.3%	1.9%	3.8%	13.0%	-6.6%	1.00
2024	-1.2%	0.6%	-2.2%	1.4%	-4.6%	1.2%	-2.1%	0.6%	-2.3%	-1.7%	-0.8%	-8.1%	-17.1%	-18.2%	-1.46
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual	Max DD	Sharpe

Buy & Hold

2015	1.1%	0.0%	-1.5%	0.6%	-0.2%	-1.5%	-0.8%	-1.4%	-0.9%	2.5%	-1.6%	-1.3%	-5.0%	-6.7%	-0.91
2016	-0.2%	1.5%	1.8%	0.9%	0.1%	2.4%	1.2%	-0.5%	0.5%	-1.6%	-2.0%	0.4%	4.8%	-4.6%	1.01
2017	1.6%	1.8%	-0.3%	0.8%	1.0%	-0.8%	1.6%	1.5%	-0.6%	1.4%	1.0%	0.9%	10.1%	-2.0%	2.91
2018	2.5%	-1.8%	-0.5%	0.0%	1.7%	-0.4%	0.3%	1.7%	-0.2%	-3.5%	-0.5%	-2.3%	-3.7%	-8.9%	-0.50
2019	3.8%	1.0%	1.6%	1.4%	-2.0%	3.9%	0.6%	1.5%	-0.5%	1.5%	0.8%	1.2%	15.7%	-2.3%	3.08
2020	1.7%	-2.0%	-3.5%	5.5%	2.7%	2.2%	5.0%	3.1%	-2.8%	-2.0%	4.4%	2.8%	16.8%	-13.1%	1.31
2021	-0.8%	-0.5%	0.1%	3.9%	1.0%	1.8%	2.1%	1.5%	-2.9%	4.2%	-0.1%	1.7%	12.8%	-5.0%	1.52
2022	-3.6%	-0.6%	2.1%	-5.9%	-0.4%	-5.0%	4.3%	-3.1%	-6.4%	2.1%	4.1%	-3.9%	-15.8%	-17.9%	-1.12
2023	5.3%	-2.3%	4.6%	0.6%	1.0%	2.7%	2.6%	-1.1%	-3.5%	-0.6%	5.5%	1.7%	17.5%	-5.8%	2.11
2024	0.8%	2.3%	2.5%	-2.1%	3.3%	2.8%	0.4%	1.1%	2.1%	-0.1%	2.3%	-1.7%	15.3%	-6.4%	1.65
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual	Max DD	Sharpe

Cost Sensitivity

Sharpe vs half-spread per fill. Commission held fixed at 0.50 bp per fill on every rung, so the whole curve already pays it.



Backtest Configuration

Field	Value
name	Score Scaling Neutral
data	data/raw
tickers	['SPY', 'QQQ', 'TLT', 'IEF', 'GLD', 'DBC', 'UUP', 'FXE']
portfolio	score_proportional
window	20
initial_cash	1,000,000.0
entry_threshold	0.0
exit_threshold	0.0
vol_window	20
max_gross	2.0
dollar_neutral	True
winsor_limit	3.0
cost_bps	1.0
commission_bps	0.5
drift_band	0.005
stop_loss_pct	None
take_profit_pct	None
max_holding_days	None
output_dir	output